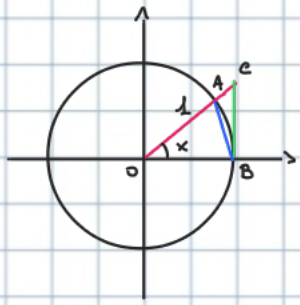
$$\lim_{x\to 0} \frac{\sin x}{x} = 1$$

1) $f(x) = \frac{\sin x}{x}$ è pari $\quad f(-x) = f(x) \Rightarrow \frac{\sin(-x)}{-x} = \frac{\sin x}{x}$

$\Rightarrow \frac{-\sin x}{-x} = \frac{\sin x}{x}$

$\lim_{x\to 0^+} f(x) = \lim_{x\to 0^-} f(x)$ $\quad \Rightarrow \frac{\sin x}{x} = \frac{\sin x}{x}$

2)



$A_{A\hat{O}B} < A_{A\hat{O}B} < A_{O\hat{C}B}$

$\frac{1}{2} OA \cdot OB \cdot \sin x < \frac{1}{2} OA^2 \cdot x < \frac{1}{2} OB \cdot \tan x \cdot OA$

$\sin x < x < \tan x$

$\sin x < x < \frac{\sin x}{\cos x}$

$1 < \frac{x}{\sin x} < \frac{1}{\cos x}$

$\cos x < \frac{\sin x}{x} < 1$

Passo al limite

$$\lim_{x\to 0} \cos x < \lim_{x\to 0} \frac{\sin x}{x} < 1$$

Per il thm del confronto $\lim_{x\to 0} \frac{\sin x}{x} = 1$

Da queste ipotesi

$$\lim_{x\to 0} \frac{1-\cos x}{x^2} = \frac{1}{2}$$

$$\lim_{x\to 0} \frac{1-\cos x}{x^2} \cdot \frac{1+\cos x}{1+\cos x} = \frac{1-\cos^2 x}{x^2} \cdot \frac{1}{1+\cos x} = \frac{1}{2}$$

$\frac{1-\cos^2 x}{x^2} \to \frac{\sin^2 x}{x} = 1^2 = 1$

$\frac{1}{1+\cos x} \to \frac{1}{2}$